

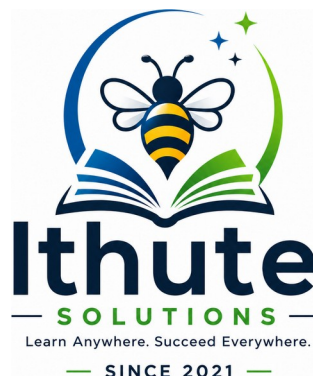


FULL USER & OPERATIONS MANUAL

Role-by-role guide for the multi-tenant Lesotho loan marketplace

Final engineered edition

Covers tenant administration, lending, realtime notifications, encrypted-at-rest chat and files, voice notes, accounting, performance, midnight reporting, audited role switching and production deployment.



Software developed and maintained by Ithute Solutions

Edition: July 2026 • Current migration head: f1a9c4e7b620



1. Document control and how to use this manual

Start here before using any operational module

Item	Value
Product	LoanHub multi-tenant loan management and marketplace platform
Product identity	LoanHub app icon and LoanHub horizontal logo
Developer	Ithute Solutions
Audience	Platform owner, company teams, branch teams, specialist roles and borrowers
Timezone	Africa/Maseru for midnight reconciliation and generated periods
Security position	HTTPS/WSS in transit; AES-GCM encryption at rest for chat and managed files
Important limitation	Encryption is server-managed, not client-to-client end-to-end encryption
Migration head	f1a9c4e7b620

How to use the guide

- Read Chapters 2–9 for product identity, account access, themes, notifications and role switching.
- Read the page for your assigned role before performing daily work.
- Use workflow chapters for marketplace, loans, chat, files, accounting and reports.
- Use the operational checklists and troubleshooting pages during daily support.
- Company policy, consumer-protection rules and Lesotho regulatory requirements override examples in this manual.

Support rule

Use friendly record references and the request ID shown by the interface. Never send passwords, mobile-money PINs, private keys, database passwords or complete tokens.



2. Product overview

How LoanHub connects borrowers, lenders and the platform owner

LoanHub allows many independent loan companies to operate in one service while company and branch data remains isolated by server-enforced membership. Borrowers publish requests, eligible lenders compete with offers, the borrower accepts one offer, and the selected company services the loan.

Core business flow

- 1.A borrower registers and completes identity, address and employment details.
- 2.The borrower broadcasts a consented loan request to the marketplace.
- 3.Companies receive a redacted request and unlock full details according to plan rules.
- 4.Authorised loan staff perform affordability and risk review and submit one company offer.
- 5.The borrower compares amount, fees, interest, term and total repayment and accepts one offer.
- 6.The selected company approves, disburses, schedules, collects and reconciles the loan.
- 7.Notifications, audit events, accounting and reports provide oversight throughout the lifecycle.

Module	Purpose
Tenant administration	Companies, branches, roles, staff and company settings
Marketplace	Broadcast requests, unlocks, offers and selection
Servicing	Loans, disbursement, schedules, repayments and collections
Transparency	Notifications, audits, errors and action links
Business operations	Employees, performance, accounting, chat, files and reports



3. Brand identity and logo use

LoanHub is the product; Ithute Solutions is the developer

Asset	Correct use	Do not use it as
LoanHub app icon	Browser/app icon, compact sidebar, mobile launcher and small report mark	The Ithute Solutions company logo
LoanHub horizontal logo	Login page, public marketing, product headers and report covers	A lender-company logo
Ithute Solutions logo	Small “developed by” credit, engineering documentation and maintenance reports	The primary LoanHub system identity

Inside the system

- The navbar and authentication screens identify the system as LoanHub.
- Generated LoanHub reports use the product logo first and a smaller Ithute Solutions developer credit.
- Loan-company tenants may later add approved company branding without replacing platform ownership marks.
- Product assets are stored in `brand/`, frontend `public/` and backend `assets/` with clear filenames.

Marketing consistency

Always describe the product as “LoanHub — Lesotho Loan Marketplace” and the engineering credit as “Developed and maintained by Ithute Solutions.”



4. Accounts, tenants and branches

Understand where permissions come from

Account

A user account stores authentication and personal identity. The account may also have a borrower profile, one or more company-staff memberships, or the platform-owner role.

Tenant

A tenant is a loan company. Company information is isolated from unrelated companies. Company IDs supplied by a browser are never trusted without checking the authenticated membership in PostgreSQL.

Branch scope

A staff membership may be assigned to a branch. Branch users receive branch-scoped lists, performance, reports, files and notifications unless a company-wide role authorises broader access.

Scope	Typical viewers	Examples
Platform	Platform owner	All companies, plans, platform accounting, critical errors
Company	Owner/admin and authorised specialists	Company portfolio, staff, reports, accounting
Branch	Branch manager and branch-assigned staff	Branch loans, employees, files and performance
Personal	Borrower or employee subject	Own requests, loans, goals, reviews and private files

Always check context

Before changing data, confirm the active company, branch and role displayed by the interface.



5. Sign-in, password recovery and session safety

Access LoanHub without exposing credentials

8. Open the official HTTPS LoanHub address and enter the registered phone number and password.
9. LoanHub redirects to the platform, company or borrower portal according to authorised access.
10. Use the theme control to select Light, Dark or System without changing permissions.
11. Use Logout on shared devices and close the browser after sensitive finance or owner work.
12. Use "Forgot password?" to submit a privacy-safe recovery request to the platform owner. The page never confirms whether an account exists.

Password recovery

The recovery page creates a friendly RST reference and sends a private notification to platform support when a matching active account exists. An authorised platform owner verifies identity through approved channels. LoanHub never sends the existing password.

- Use a long, unique passphrase.
- Never send passwords, tokens, JWT keys or mobile-money PINs in chat.
- Do not save owner or finance passwords on shared computers.
- Unexpected tenant or role access must be reported immediately.



6. Navigation, themes and accessible use

Use every workspace comfortably on desktop and mobile

Navigation

- The sidebar contains only modules allowed by the active role.
- The notification bell opens a recent inbox and shows unread count.
- The floating chat button opens the correct chat route for the current portal.
- Search, filters, pagination and responsive cards reduce large-table overload.
- Mobile navigation preserves the same permission rules as desktop navigation.

Theme control

Select Light, Dark or System from the header theme control. System follows the device preference. Theme selection changes presentation only; it never changes access, reporting or stored business data.

Human-friendly references

Record	Example	Use when contacting support
Payment	PAY-2026-8A74B113	Reference, company and date
Branch	BRN-2026-14CC90F2	Branch name and reference
Report	RPT-20260715-7C42A1D9	Title, period and reference
Password recovery	RST-20260715-2C8D91F0	Recovery reference only
Error	Request ID shown by the safe error message	Page, time, role and request ID



7. Notifications and action routing

Receive useful events without notification spam

13. Open the navbar bell to view the latest notification cards.
14. Unread count increases when a role-relevant event is persisted.
15. Select a notification to mark it read and navigate to the linked action page.
16. Use the full notification centre to search, filter, archive or mark all as read.
17. WebSocket delivery is instant; PostgreSQL persistence means a missed socket message still appears after refresh.

What enters the inbox

Actionable grouped events enter the inbox. Detailed CRUD changes remain in the audit trail. Large operations such as creating a repayment schedule may create many audit records but only one useful notification.

Event	Recipients	Action
New company registration	Platform owner	Review company application
New marketplace request	Eligible lender roles	Review redacted request
Offer received	Borrower	Compare offer
Failed payment	Relevant finance/collections and affected party	Inspect transaction
Critical unhandled error	Platform owner only	Open system-error incident



8. Safe error handling and technical incidents

Ordinary users see safe help; platform owners see diagnostics

Validation and permission problems are displayed as readable messages. An unhandled backend failure is assigned a request ID, stored in the incident log, grouped by fingerprint and broadcast only to active platform owners.

What ordinary users see

Safe message

"The action could not be completed. The platform owner has been notified." The message includes a request ID, but never a stack trace, SQL statement, secret or internal file path.

What the platform owner sees

- Endpoint, method, status, request ID and fingerprint.
- Occurrence count and first/last seen times.
- Safe affected user/company/branch context.
- Full diagnostic message and stack trace on the restricted incident page.
- Resolve, document and reopen controls.

Spam control

Recurring fingerprints are grouped and technical notifications are suppressed for a short window. The occurrence counter still increases, preserving transparency without overwhelming the owner inbox.



9. Platform-owner role switching

Test and support roles without creating many browser sessions

18. Select Switch role from the platform-owner header.
19. Search an active borrower or company membership by name, company, branch or role.
20. Enter a clear support/testing reason.
21. Start the time-limited session. An amber banner remains visible at all times.
22. Perform the required workflow test, then select Return to restore the preserved platform-owner session.

Safeguards

- Only a platform owner can start role switching.
- Another platform-owner account cannot be impersonated.
- The target company membership is verified by the backend.
- The token contains the original actor and expires after the configured limit.
- The start event and all actions retain impersonation context in the audit trail.
- If the impersonated token expires, the frontend restores the original owner session rather than attempting an ordinary refresh.

Not an anonymity feature

Role switching is for support and workflow testing. It must never be used to hide the responsible platform owner.



10. Platform owner / SuperAdmin

Govern the complete multi-tenant service, commercial model, security and technical operations.

What this role can do

- Approve, reject, suspend and activate loan companies.
- Configure subscription plans, unlock fees, transaction fees, limits and feature switches.
- Review platform companies, loans, payments, accounting, reports, performance, audit and critical errors.
- Use audited role switching to test role workflows.
- Communicate with authorised users through platform chat and managed files.

Recommended daily routine

23. Review pending companies, high-priority notifications and system errors.
24. Check payment, loan and subscription exceptions.
25. Review audit activity and unusual role or tenant changes.
26. Review platform accounting and generated reconciliation reports.
27. Follow up unresolved support conversations.

Control boundaries

- Do not permanently delete financial history.
- Protect JWT keys, encryption keys and provider credentials outside Git.
- Use role switching only with a documented reason.
- Run migrations only after backup and staging validation.

Main workspaces

Workspace	Purpose
/superadmin	Live platform dashboard
/superadmin/companies	Tenant applications and company controls
/superadmin/plans	Commercial plan configuration
/superadmin/system-errors	Restricted technical incidents
/superadmin/accounting	Platform accounting
/superadmin/reports	Platform/company/branch reports



11. Company owner

Control one loan company as the highest tenant authority.

What this role can do

- Manage legal settings, branches, staff memberships, roles and loan products.
- Oversee marketplace, offers, loans, payments and collections.
- Control subscription choices and company-wide limits.
- Access accounting, reports, employees, performance, files, chat and audit activity.

Recommended daily routine

28. Confirm company approval and subscription status.
29. Review active, overdue and failed operational items.
30. Resolve staffing, branch and access exceptions.
31. Review accounting balances and daily reconciliation reports.
32. Review company performance and outstanding support cases.

Control boundaries

- Assign least privilege and separate finance, lending, collections and audit duties.
- Do not edit history to hide corrections; use controlled entries and audit notes.
- Keep licence and company contact details current.

Main workspaces

Workspace	Purpose
/company	Company dashboard
/company/settings	Legal and operating profile
/company/branches	Branch structure
/company/staff	Staff and roles
/company/accounting	Company books
/company/reports	Operational and financial reporting



12. Company administrator

Run day-to-day company administration under the company owner.

What this role can do

- Maintain company master data, branches, staff and products.
- Coordinate lending, finance, collections, support and HR workflows.
- Review company notifications, chat, files, performance and reports.

Recommended daily routine

33. Prioritise notifications by severity and due date.
34. Check incomplete records and inactive dependencies.
35. Route work to the correct specialist role.
36. Review audit activity for unexpected changes.

Control boundaries

- Do not grant administrator access to solve a temporary task.
- Confirm branch assignment before activating membership.
- Use managed group chat instead of external uncontrolled channels.

Main workspaces

Workspace	Purpose
/company	Operational overview
/company/notifications	Actionable events
/company/activity	Company audit history
/company/chat	Operational communication



13. Branch manager

Manage staff, lending and performance for an assigned branch.

What this role can do

- View branch-scoped loans, employees, files, reports and performance.
- Coordinate loan officers, finance and collections.
- Escalate risk, compliance and technical issues.

Recommended daily routine

37. Review branch queue, staff workload and overdue loans.
38. Assign work and confirm hand-offs.
39. Review branch performance and reconciliation report.
40. Escalate exceptions with supporting references.

Control boundaries

- Do not access another branch without company-wide authority.
- Confirm customer identity before sensitive actions.
- Report conflicts of interest and suspected fraud.

Main workspaces

Workspace	Purpose
/company/loans	Branch-permitted loans
/company/performance	Branch analytics
/company/reports	Branch reports
/company/chat	Branch communication



14. Loan officer

Evaluate borrower requests and prepare responsible, competitive offers.

What this role can do

- Review redacted marketplace requests.
- Unlock borrower details under authorised billing rules.
- Assess affordability and product fit.
- Create and track company offers.

Recommended daily routine

41. Review new and expiring requests.
42. Check documents, consent, income and obligations.
43. Prepare a compliant offer using an active product.
44. Monitor accepted, rejected and expired outcomes.

Control boundaries

- Do not disclose borrower details to another company.
- Do not promise approval before required checks.
- Do not enter rates outside authorised product and policy limits.

Main workspaces

Workspace	Purpose
/company/marketplace	Broadcast requests
/company/products	Approved product rules
/company/loans	Selected/serviced loans
/company/chat	Borrower communication



15. Finance officer

Control disbursements, receipts, reconciliation and accounting evidence.

What this role can do

- Review and initiate authorised disbursement and repayment transactions.
- Reconcile provider references, LoanHub ledger and accounting entries.
- Review trial balance, profit and loss and balance sheet.

Recommended daily routine

45. Review pending and failed payments.
46. Verify amount, payer/payee and authorisation.
47. Reconcile successful provider result to LoanHub records.
48. Post or review related journal evidence.

Control boundaries

- A pending database payment is not proof of provider settlement.
- Never mark success without verified evidence.
- Keep provider credentials and callbacks restricted.

Main workspaces

Workspace	Purpose
/company/payments	Company payment ledger
/company/accounting	Journals and statements
/company/reports	Financial reports
/company/files	Payment evidence



16. Collections officer

Follow due and overdue repayments professionally and transparently.

What this role can do

- Review assigned or branch overdue loans and schedules.
- Inspect payment history before contacting borrowers.
- Record follow-up, promises, disputes and escalation.

Recommended daily routine

49. Prioritise overdue queue by age, amount and risk policy.
50. Confirm no completed payment is awaiting reconciliation.
51. Contact borrower through approved channel.
52. Update outcome and next action.

Control boundaries

- Follow respectful-contact and consumer-protection policy.
- Do not expose debt in group chat.
- Do not delete failed or reversed payment evidence.

Main workspaces

Workspace	Purpose
/company/loans	Balances and schedules
/company/payments	Payment evidence
/company/chat	Private follow-up



17. Compliance officer

Monitor consent, access, regulatory controls and remediation.

What this role can do

- Review audit activity, role changes, company records and high-risk notifications.
- Collect controlled evidence from reports and files.
- Document and monitor remediation.

Recommended daily routine

53. Review high-risk events and changes.
54. Filter audit by actor, entity, action, company and branch.
55. Attach evidence with confidential visibility.
56. Escalate and track remediation owners.

Control boundaries

- Do not alter evidence under review.
- Limit audit exports and personal information.
- Apply legal retention and privacy advice.

Main workspaces

Workspace	Purpose
/company/activity	Audit transparency
/company/files	Controlled evidence
/company/reports	Compliance reports



18. Auditor

Independently review operational, financial and access evidence.

What this role can do

- Use read-oriented activity, accounting, reports and performance.
- Extract evidence with friendly references.
- Record findings according to approved methodology.

Recommended daily routine

57. Define period, branch/company and process scope.
58. Compare approvals, source events, payments and postings.
59. Test samples and record findings.
60. Follow up management response.

Control boundaries

- Maintain independence and avoid operational posting.
- Raw UUIDs are diagnostics, not primary evidence labels.
- Protect exported evidence.

Main workspaces

Workspace	Purpose
/company/activity	Source events
/company/accounting	Financial evidence
/company/reports	Period reports



19. Customer support

Resolve account, navigation and workflow questions without exceeding permission.

What this role can do

- Use role-filtered chat, notifications and safe record context.
- Guide users to correct pages and statuses.
- Escalate finance, compliance, lending or technical cases.

Recommended daily routine

61. Review unread conversations.
62. Verify the user through approved questions.
63. Ask for a friendly reference and request ID.
64. Give safe guidance or escalate.

Control boundaries

- Never ask for a password, PIN, private key or full token.
- Do not reveal data to an unauthorised person.
- Use direct chat for personal cases.

Main workspaces

Workspace	Purpose
/company/chat	Support conversations
/company/notifications	Action alerts
/company/files	Safe exchange



20. HR manager

Maintain protected workforce records and employment lifecycle.

What this role can do

- Create and update employee profiles, titles, departments, managers and dates.
- Coordinate goals and reviews.
- Manage confidential HR files and authorised salary fields.

Recommended daily routine

65. Review hires, probation, transfers and terminations.
66. Keep reporting lines and employee numbers accurate.
67. Coordinate access deactivation with employment changes.
68. Review overdue goals and reviews.

Control boundaries

- Do not place salary, medical or disciplinary data in ordinary group chat.
- Use confidential visibility for protected HR files.
- Employment status and system access are related but distinct.

Main workspaces

Workspace	Purpose
/company/employees	Employee records
/company/performance	Goals and reviews
/company/files	HR documents



21. Performance manager

Set measurable goals and maintain fair, evidence-based reviews.

What this role can do

- Create goals, update progress and perform formal reviews.
- Compare individual, branch and company trends.
- Use reports and audit evidence to validate metrics.

Recommended daily routine

69. Review goals due or overdue.
70. Check data quality and workload context.
71. Prepare and complete evidence-based reviews.
72. Discuss results with employee and manager.

Control boundaries

- Do not use raw activity volume as the only measure.
- Investigate missing data before penalising an employee.
- Account for portfolio and branch differences.

Main workspaces

Workspace	Purpose
/company/performance	Performance dashboard
/company/employees	Employee context
/company/reports	Period evidence



22. Risk manager

Identify credit, operational, payment and fraud risk.

What this role can do

- Review overdue rates, failed payments, concentrations and unusual activity.
- Use audit events, reports and files to investigate.
- Recommend and monitor controls.

Recommended daily routine

73. Review high-risk notifications and trends.
74. Investigate unusual transactions or access changes.
75. Assign mitigation owner and due date.
76. Verify outcome in the next reporting period.

Control boundaries

- Separate facts from inferred risk.
- Do not export unnecessary personal data.
- Escalate suspected fraud immediately.

Main workspaces

Workspace	Purpose
/company/reports	Risk indicators
/company/activity	Event evidence
/company/performance	Operational trends



23. IT support

Diagnose incidents and assist users without bypassing business controls.

What this role can do

- Receive platform-owner-assigned technical context.
- Use request IDs, fingerprints and safe logs.
- Test fixes in staging and document resolution.

Recommended daily routine

77. Review assigned incident and occurrence trend.
78. Reproduce with non-sensitive data.
79. Patch, validate and deploy through controlled workflow.
80. Monitor recurrence after resolution.

Control boundaries

- Ordinary tenant users do not receive stack traces.
- Do not edit financial data to make an error disappear.
- Never store secrets in Git or chat.

Main workspaces

Workspace	Purpose
/superadmin/system-errors	Restricted incidents
/superadmin/activity	Technical audit context
/superadmin/chat	Secure support coordination



24. Borrower

Request funding, compare offers and manage personal loans and payments.

What this role can do

- Maintain own profile, requests, offers, loans, repayments, files and chat.
- Compare offers and accept one.
- Track disbursement, schedule, balance and payment result.

Recommended daily routine

81. Keep contact, identity, address and employment data current.
82. Review new offers and expiry times.
83. Check upcoming instalments and payment status.
84. Report incorrect balances or unrecognised transactions.

Control boundaries

- Never pay a person outside the approved provider/reference flow.
- Never send a mobile-money PIN or password in chat.
- Use private conversation/file visibility for personal documents.

Main workspaces

Workspace	Purpose
/borrower/requests	Requests and offers
/borrower/loans	Accepted loans
/borrower/payments	Repayments
/borrower/chat	Company/support communication
/borrower/files	Personal/shared files



25. Company registration and approval

Create a tenant and first owner account

The registration page separates company identity, operations, owner details and final review. The left panel explains the owner role while the right panel shows step progress.

Workflow

85. Choose Loan company registration.
86. Enter legal company name, registration/licence references and company phone.
87. Enter email, website, district and business address.
88. Create the first owner identity and password.
89. Review the declaration and submit.
90. Platform owner verifies evidence and approves, rejects or returns the application.

Important controls

- The company starts pending. Lending remains restricted until approval and applicable billing conditions are satisfied.
- Passwords are never saved in the registration draft.

26. Company settings, branches and staff

Build a controlled operating structure

Company settings contain legal, contact and location sections with validation, profile-completeness tracking and read-only behaviour for unauthorised staff.

Workflow

91. Company owner/admin opens Company settings and updates official data.
92. Create branches with name, district, town, address, phone and email.
93. Create or select a user membership.
94. Assign one approved company role and optional branch.
95. Activate the membership and ask the user to confirm scope.
96. Reassign dependencies before deactivating a branch.

Important controls

- Do not delete branches or companies that have financial history.
- Use company/branch access checks on every backend endpoint.
- Review access when an employee transfers or leaves.

27. Employee and performance management

Connect people records to measurable work

Employee profiles extend staff membership with job, department, manager, employment dates, skills, targets, protected salary data and HR notes.

Workflow

97. Create the employee profile from an active staff membership.
98. Assign employee number, branch, job title, department and manager.
99. Record employment type, status and dates.
100. Create measurable goals with period, target and owner.
101. Update progress and attach evidence.
102. Complete formal review and notify the employee/manager.

Important controls

- Salary and confidential HR notes are visible only to authorised management/HR roles.
- Use evidence and operating context; do not rank staff only by activity count.



28. Loan products and marketplace requests

Use controlled rules before competing for a borrower

Loan products define amount, term, interest, fee and availability boundaries. Marketplace request cards remain redacted until the company has authorised access.

Workflow

103. Company management creates and activates approved products.
104. Borrower creates a request with amount, term, purpose and consent.
105. LoanHub broadcasts the request to eligible active companies.
106. Loan staff review the redacted request and product fit.
107. The company unlocks full details through subscription or pay-per-request access.
108. Authorised staff prepare one compliant company offer.

Important controls

- Do not expose borrower identity before a valid unlock.
- Do not pay an unlock fee unless authorised.
- Use product rules and documented risk policy.



29. Offers, selection and loan creation

Move from competition to one selected agreement

An offer records amount, interest, term, fees, estimated instalment, total repayment, expiry and conditions.

Workflow

109. Borrower opens the request and compares all active offers.
110. Review total repayment and conditions, not only the instalment.
111. Borrower accepts one preferred offer.
112. LoanHub locks the selection within a transaction.
113. Competing active offers are rejected.
114. The chosen company completes approval and servicing.

Important controls

- One company should have only one active offer per request.
- Acceptance must be atomic to prevent double selection.
- The accepted offer and resulting loan remain fully auditable.

30. Disbursement and repayment schedules

Move approved value safely and create due dates

The selected company verifies the agreement and payment details before creating an outbound disbursement. A successful provider result activates servicing and the repayment schedule.

Workflow

115. Verify borrower payee details and authorised amount.
116. Initiate the approved provider transaction.
117. Wait for verified success or approved mock completion in development.
118. Create the schedule according to product/offer terms.
119. Review first due date, instalments and maturity.
120. Notify borrower and relevant company roles.

Important controls

- A pending transaction is not settlement.
- Live M-Pesa/EcoCash requires official callback authentication, status enquiry and reconciliation.
- Schedule generation is grouped in the inbox but detailed in the audit trail.

31. Repayments, allocations and collections

Apply money to the correct loan and instalment

Repayment creates an inbound transaction. After verified success, the amount is allocated to instalments, paid values and balance are updated, and accounting hooks can create the related journal evidence.

Workflow

121. Borrower or authorised staff initiates repayment.
122. Provider confirms success or approved mock mode completes.
123. Allocate payment to due instalments according to the servicing rule.
124. Update amount paid, balance and loan status.
125. Reconcile payment, allocation and accounting entry.
126. Collections verifies history before follow-up on overdue items.

Important controls

- Use idempotency keys to prevent duplicate payment creation.
- Never delete reversals or failed-payment evidence.
- Record borrower disputes and promises transparently.



32. Billing, subscriptions and plan controls

Configure how companies pay for LoanHub

The platform owner manually controls plan price, annual price, unlock fee, transaction percentage, limits, feature flags, visibility and active status.

Workflow

127. Platform owner creates or edits a plan.
128. Company owner selects monthly, annual or pay-per-transaction cycle.
129. A zero-price plan activates without a useless payment record.
130. A paid plan creates a payment and remains pending until confirmed.
131. After activation, other active subscriptions are cancelled.
132. Maintenance expires subscriptions at the correct local date.

Important controls

- STARTER remains the fallback code but its commercial values are editable.
- LoanHub transaction fees are separate from provider charges.
- Used plans should be made private/inactive rather than deleted.

Billing options

Option	Behaviour
Monthly	Configured monthly price and term
Annual	Configured annual price and annual term
Pay per request	Configured marketplace unlock fee
Transaction fee	Platform percentage under the commercial model



33. WhatsApp-style platform chat

Communicate across authorised platform relationships

LoanHub chat provides direct and group conversations inside the platform. It is not Meta WhatsApp. Directory results are filtered by role, tenant relationship and active membership.

Workflow

133. Open Chat or the floating chat button.
134. Create a direct chat with one authorised user or a group with several users.
135. Send text, PDFs, images, office files or voice notes.
136. Watch online/offline and last-seen status.
137. Use typing and read indicators to coordinate.
138. Open shared attachments through authenticated file downloads.

Important controls

- Borrowers can contact relevant lenders and platform support, not unrelated companies.
- Do not share passwords, PINs or private keys.
- Sensitive borrower or HR information belongs in a direct conversation.



34. Chat presence, voice notes and encryption

Understand what is protected and how presence works

Authenticated WebSocket connections drive realtime message, typing, notification and presence events. Redis distributes events across API workers, while PostgreSQL remains the durable source of truth.

Workflow

139. Grant microphone permission only on the official HTTPS domain.
140. Record a voice note, review it and send it as a managed audio file.
141. Online means at least one authenticated socket is connected.
142. Offline users show the last saved activity time when available.
143. If the socket disconnects, the provider reconnects and polling preserves durable data.

Important controls

- Text and file bytes use authenticated AES-GCM encryption at rest.
- HTTPS/WSS protects traffic in transit.
- This is not end-to-end encryption: the authorised LoanHub server can decrypt content for access-controlled delivery.
- Encryption keys must be backed up securely; losing them can make encrypted content unreadable.

Security language

Market the feature as “encrypted in transit and encrypted at rest,” not as client-to-client end-to-end encryption.



35. File centre and PDF handling

Store, share, download and print documents safely

One managed-file layer supports chat attachments, generated reports, company documents, borrower files, HR/compliance evidence and accounting support.

Workflow

144. Select Upload or attach a file in chat.
145. Choose the narrowest visibility: private, conversation, branch, company, confidential or platform.
146. The server validates size, filename, declared type, detected signature and allowed extension.
147. File bytes are encrypted, checksummed and written atomically.
148. Authorised users download through a streamed endpoint with a safe filename.
149. Generated reports appear in both Reports and Files and can be downloaded for printing.

Important controls

- Executable and suspicious mismatched content is blocked or quarantined.
- Back up database metadata and encrypted file storage together.
- For multiple servers, use S3/MinIO-compatible object storage and malware scanning.

Visibility

Level	Who can access
Private	Owner/uploader
Conversation	Active conversation participants
Branch	Branch users and authorised management
Company	Active members of the tenant
Confidential	Authorised management/platform owner
Platform	Platform owner



36. Accounting

Use the double-entry foundation responsibly

Accounting supports platform and company ledgers, chart of accounts, balanced journals, branch allocation, evidence references and core statements.

Workflow

150. Initialise the default chart when required.
151. Review or add approved custom accounts with unique codes.
152. Create a journal date, description and business reference.
153. Add at least two lines and ensure debit equals credit.
154. Save as draft, review evidence and post.
155. Run trial balance, profit and loss and balance sheet.

Important controls

- Posted entries should be corrected through controlled entries rather than hidden deletion.
- A qualified accountant must approve the chart, mappings, provider fees, tax, bad debt and statutory treatment.



37. Manual and scheduled reports

Generate operational and financial evidence

Authorised users can generate PDF or CSV reports for branch, company or platform scope and select a period. The report uses LoanHub branding with a smaller Ithute Solutions developer credit.

Workflow

156. Open Reports.
157. Choose report type, scope, output format and period.
158. Select authorised company/branch context.
159. Generate and wait for completed status.
160. Download from Reports or Files.
161. Create a recurring schedule when ongoing generation is required.

Important controls

- Report access follows tenant and branch scope.
- Use friendly report references in support and audit communication.
- Recipient email storage does not mean automatic email delivery unless a mail adapter is configured.

38. Midnight reconciliation and automatic reports

What happens at 00:00 Africa/Maseru

The maintenance service calculates local Lesotho midnight independently from the container timezone. A PostgreSQL advisory lock ensures only one worker performs the nightly cycle.

Workflow

162. At local 00:00, run lifecycle maintenance for requests, offers, subscriptions, instalments and loans.
163. Generate platform daily PDF and CSV for the previous day.
164. Generate one report for every active company.
165. Generate one report for every active branch.
166. On Monday, add the previous completed week.
167. On the first day of a month, add the previous completed month.
168. On January 1, add the previous calendar year.

Important controls

- Generation is idempotent: the same scope, period, type and format is not duplicated.
- Reports are saved in managed media and are ready for authorised download or printing.
- The maintenance container must remain healthy.



39. Audit history and transparency

Trace who changed what, where and when

Normal mapped create, update and delete events create durable audit records with actor, role, tenant, branch, entity, before/after values, changed fields, request ID, client context and time. Secrets are redacted.

Workflow

169. Open Activity.
170. Filter by period, actor, action, entity, company or branch.
171. Open an event and compare before/after values.
172. Use the request ID to correlate an incident.
173. Export or attach evidence under authorised visibility.

Important controls

- Bulk business workflows use mapped changes so audit listeners are not bypassed.
- Role-switch actions retain the original platform owner in event context.
- Audit records are not a substitute for approved legal retention policy.



40. Dashboard quality and readable records

Avoid demo data and raw database identifiers

Dashboards use live provider data, meaningful names and compact references. The old static company-admin dashboard is retained only as a redirect to the live company portal.

Workflow

- 174. Confirm a card is calculated from AppDataProvider or an authorised API response.
- 175. Display company, branch, borrower, employee and loan names where permitted.
- 176. Use friendly record references for payments, reports and support requests.
- 177. Keep raw UUIDs in technical details only.
- 178. Make empty, loading and error states explicit.

Important controls

- Do not show invented totals, fake recent activity or inactive buttons.
- Buttons either submit a validated form, open a menu, navigate to a real route or are clearly disabled with a reason.



41. Security, privacy and data handling

Operate LoanHub responsibly

Security depends on server-side tenant enforcement, least privilege, encrypted secrets, safe files, controlled deployments and honest provider status.

Workflow

179. Use HTTPS/WSS and secure cookies/tokens in production.
180. Keep JWT, encryption, database and mobile-money secrets outside Git.
181. Back up PostgreSQL and encrypted file storage together and test restoration.
182. Review roles regularly and remove access promptly.
183. Protect borrower consent and minimise exported personal data.
184. Use the request ID rather than technical stack traces for ordinary support.

Important controls

- Live M-Pesa/EcoCash remains incomplete until official provider onboarding and certification.
- Do not describe mock or pending adapters as live settlement.
- Use signed object-storage links and malware scanning for high-availability deployment.



42. Database migrations and release validation

Keep models, Alembic and routes aligned

The repository has one repaired linear migration chain ending at ``f1a9c4e7b620``. Compatibility revisions remain because production/local databases may already reference them.

Workflow

185. Back up the target database.
186. Stop API, maintenance and other sessions that can lock altered tables.
187. Run ``alembic current``, ``alembic heads`` and inspect the graph.
188. Use the dedicated migration container or guarded migration script.
189. Run ``alembic upgrade head``.
190. Run ``alembic check`` against staging and execute release validation.
191. Start services and smoke-test all roles.

Important controls

- Never delete an applied migration file.
- Never use ``alembic stamp head`` to hide missing schema changes.
- Do not create meaningless “test” autogenerated revisions.
- Review every generated migration before applying it.

Validation

Check	Expected
Alembic heads	Exactly one: <code>f1a9c4e7b620</code>
SQLAlchemy	All mappers configure; audit fields remain scalar columns
Routes	No missing router, duplicate method/path or duplicate operation ID
Frontend	Typecheck, lint and production build pass
Docker	<code>`docker compose config`</code> passes and health checks succeed



43. One-stack Docker architecture

Run frontend and backend together

The root `compose.yaml` is the production entry point. The two codebases remain separated under `apps/` but deploy as one managed stack.

Workflow

192. PostgreSQL stores durable business data.
193. Redis distributes realtime notifications, presence and chat across workers.
194. The migrate service applies Alembic before API startup.
195. FastAPI serves `/api/v1`, `/health` and WebSockets.
196. The maintenance service reconciles lifecycle and midnight reports.
197. Next.js serves the user interface.
198. Caddy provides HTTPS, WSS, compression, limits and security headers.

Important controls

- Only Caddy exposes ports 80/443.
- Private JWT files are mounted read-only from `secrets/`.
- Media uses a persistent volume.
- Do not run migrations independently in every API worker.

44. GitHub to Hostinger deployment

Deploy a pushed main branch through CI/CD

The production workflow validates the source in GitHub Actions and then connects to the Hostinger VPS over SSH. The VPS script fetches the exact committed main branch, creates a database backup when available, builds images, migrates once and restarts the complete Compose stack.

Workflow

199. Create a Hostinger VPS and install Docker/Compose.
200. Clone the repository into the deployment path.
201. Create server `.env`` and JWT PEM files.
202. Add GitHub repository secrets for host, port, user, path and SSH key.
203. Push to ``main``.
204. Wait for backend, frontend and Compose validation.
205. The deploy job runs the guarded server script.
206. Verify ``https://your-domain/health`` and role smoke tests.

Important controls

- Protect the production GitHub environment with approvals.
- Use a dedicated deploy SSH key with limited VPS access.
- Keep the server `.env``, keys, backups and media outside Git.
- Review logs and rollback using the pre-migration backup if required.



45. Daily operational checklist

Minimum checks for every working day

Area	Branch/company team	Platform owner
Notifications	New requests, offers, payments, overdue items and assigned work	Pending companies, critical errors, subscription/platform exceptions
Lending	Expiring requests/offers, disbursements and schedules	Platform portfolio and unusual tenant patterns
Payments	Pending/failed provider results and reconciliation	Platform fee/billing failures and repeated incidents
Chat/support	Unread customer and internal conversations	Escalated support and cross-tenant incidents
Files	Failed/quarantined uploads and required evidence	Platform/confidential file exceptions
Accounting	Unposted journals and evidence mismatch	Platform books and cross-system reconciliation
Reports	Previous-day branch/company report available	Platform/company/branch nightly report completion

Start with exceptions

The dashboard should help users act on real exceptions, not display static demonstration cards.



46. Weekly, monthly and annual governance

Use generated periods for continuous improvement

Weekly

- Compare branch performance and collections progress.
- Review staff workload, goals and support backlog.
- Reconcile payments and inspect audit exceptions.
- Review subscription limits and file-storage health.

Monthly

- Review profit and loss, balance sheet and portfolio trend.
- Review role/access matrix and employee review completion.
- Test database and media restoration.
- Analyse recurring system-error fingerprints.

Annual

- Verify company licences and regulatory evidence.
- Review subscription/commercial model and provider contracts.
- Perform security, retention and business-continuity assessment.
- Review product, performance and accounting frameworks.

Automatic periods

Monday, month-start and January 1 midnight cycles generate the previous completed weekly, monthly and annual periods.



47. Troubleshooting

Common problems and safe actions

Problem	Likely cause	Safe action
Page unavailable	Role, tenant status or branch scope	Confirm active context and ask authorised management
Notification count delayed	Socket/network reconnect	Refresh; persisted notification remains available
Chat delayed	Socket unavailable	Wait for reconnect/polling and use stable network
Voice note fails	No HTTPS or microphone permission	Use official HTTPS domain and allow microphone
File upload fails	Size, signature, visibility or storage	Check allowed type/limit; contact support with request ID
Report missing	Maintenance stopped, scope invalid or generation failed	Generate manually and inspect worker/incident
Journal rejected	Debits and credits differ	Correct lines before posting
Payment pending	No verified provider result	Do not force success; perform authorised enquiry
Migration deadlock	API/worker holds table lock	Stop services, inspect sessions and run migrate service
Safe 500 message	Unhandled backend error	Send page, time, role and request ID to platform support



48. Status and reference glossary

Understand common lifecycle language

Area	Typical values	Meaning
Company	pending, approved, rejected, suspended	Review and operating state
Request	draft, open, offered, accepted, cancelled, expired	Marketplace lifecycle
Offer	pending, accepted, rejected, expired, withdrawn	Company proposal lifecycle
Loan	pending, approved, active, completed, defaulted, cancelled	Servicing lifecycle
Payment	pending, processing, succeeded, failed, reversed, cancelled	Transaction lifecycle
Subscription	pending, active, expired, cancelled, failed	Commercial access lifecycle
Report	queued, generating, completed, failed	Generation lifecycle
Journal	draft, posted	Accounting lifecycle
File scan	pending, clean, quarantined, failed	Managed-file safety status
Presence	online, offline, last seen	Authenticated chat availability



49. Deployment administrator checklist

Before and after a production release

Before deployment

- Review change set and migration.
- Confirm one Alembic head.
- Run backend release validation.
- Run frontend typecheck, lint and build.
- Validate Compose.
- Back up PostgreSQL and media.
- Confirm secrets and domain settings.

After deployment

- Check database, Redis, API, maintenance, frontend and Caddy health.
- Smoke-test login, tenant isolation and role switching.
- Send a chat message and voice note and download an attachment.
- Trigger a safe test report and verify branded PDF/CSV.
- Verify ordinary users cannot open technical incident details.
- Verify midnight worker timezone and next run.
- Review logs without exposing secrets.

Rollback discipline

Keep the pre-migration database backup and the previous image/release tag until production verification is complete.



50. Support, training and continuous improvement

Keep LoanHub trustworthy and marketable

LoanHub is developed and maintained by Ithute Solutions. The product should evolve through controlled releases, migration testing, user feedback, role training and transparent reports—not through hidden data changes or unsupported production experiments.

When reporting an issue

- Page or module name.
- Friendly record reference.
- Time and timezone.
- User role, company and branch.
- Safe screenshot.
- Request ID shown by the interface.
- Steps that reproduce the problem without sharing secrets.

Training approach

- Train platform owner first, then company owners/admins, then specialist and branch roles, then borrowers/support.
- Use platform-owner role switching to test role journeys with an amber audit banner.
- Use staging and non-sensitive test data for training.
- Review generated daily reports and audit history during the first production weeks.

Product statement

LoanHub is the system. Ithute Solutions is the developer and maintenance partner. Learn Anywhere. Succeed Everywhere.

